

GERMANIUM SMALL SIGNAL TRANSISTORS

PRO ELECTRON TYPES

Type	Polar- ity	V_{CB0} V Max	V_{EB0} V Max	V_{CE} V Max		I_{CB0} @ V_{CB} V	I_{CB0} μA Max	h_{FE} Min Max	@ I_C mA	C_{ob} pf Max	f_{β} MHZ Min	Pack Outline	Power Dissipation @ 25°C MW
AC107 AC116 AC117	P P P	15 30 32	12 10	18 18		6 6	8 18	35- 50-140 40-	300 ¹ 20 150	21 ²		TO-1 NS257 NS257	80 145 260
AC121-IV AC121-V AC121-VI AC121-VII	P P P P	20 20 20 20	10 10 10 10	20 20 20 20				30-60 50-100 75-150 125-250	100 100 100 100	40 40 40 40	1.5 ² 1.5 ² 1.5 ² 1.5 ²	TO-1 TO-1 TO-1 TO-1	900 900 900 900
AC122 AC122/30 AC123 AC124 AC125	P P P P P	30 45 45 45 32	18 12 12 10 10	18 32 32 32 12		6 6 6 6 10	8 8 8 18 10	40-200 40-200 50-140 40-170 50-	2 ¹ 2 ¹ 20 150 2	21 ² 21 ² 21 ²		TO-1 TO-1 NS257 NS257 TO-1	130 130 145 260 500
AC126 AC127 AC128 AC128K	P N P P	32 32 32 32	10 10 10 10	12 12 16 16		10 10 10 10	10 10 10 10	65- 50- 55-175 55-175	2 500 50 50	50 70 ² 100 100	1.7 2.5 1.0 1.0	TO-1 TO-1 TO-1 NS257	500 340 1,000 1,000
AC130 AC131 AC132 AC138 AC139	N P P P P	20 30 32 32 32	10 10 10 10 10	18 12 20 20 20		6 0.5 10 10 10	18 10 15 15 15	40- 135 ² - 30- 40-160	150 20 5 ¹ 400	40 ²	2.0 1.3	TO-1 TO-1 TO-1 TO-1 TO-1	212 500 720 720 720
AC141 AC142 AC142K AC151 AC152	N P P P P	32 32 32 32 32	10 10 10 10 10	18 20 20 2Y 2Y		10 10 10 10 0.5	35 15 15 10 10	40-160 40-160 40-160 30 ¹ - 30-150	400 400 400 2 100	27 ² 40	1.5 ² 1.5 ²	TO-1 TO-1 TO-1 NS257 TO-1	720 720 860 900 900
AC153 AC153K AC162 AC163	P P P P	32 32 32 32	10 10 10 10	18 18 2Y 2Y		10 10 10 10	10 10 10 10	50-250 50-250 100 ² 65-	300 300 50 2	100 100 40 40	1.0 1.0 1.3 1.7	TO-1 NS257 TO-1 TO-1	1,000 1,000 900 900
AC173 AC176 AC178 AC179 AC180	P N P N P	32 32 20 20 32	10 10 10 10 20	2Y 18 15 15 16		10 6 6 6	35 35 10 ²	50- 50-250 60- 60- 50-250	2 ¹ 300 150 150 600	100 ²	1.0 1.0 2.5 ²	TO-1 TO-1 NS257 NS257 TO-1	200 700 180 180 600
AC180K AC181 AC181K AC182 AC183	P N N P N	32 32 32 32 32	20 20 20 20 20	16 16 16 18 16				50-250 50-250 50-250 50- 50-	600 600 600 1 ¹ 2 ¹		2.5 ² 4.5 ² 4.5 ² 4.0 ² 4.0 ²	NS257 TO-1 NS257 TO-1 TO-1	2,500 600 2,500 200 250
AC184 AC185 AC187 AC187K AC188	P N N N P	32 32 25 25 25	20 20 10 10 10	16 16 15 15 15		10 10 10 10	35 35 15	50-250 50-250 100-500 100-500 100-500	300 300 300 300 300	180 180 110	2.5 ² 2.5 ² 1.0 1.0 1.0	TO-1 TO-1 TO-1 NS257 TO-1	600 600 1,000 1,000 1,000
AC188K AC193 AC193K AC194 AC194K	P P P N N	25 32 32 32 32	10 10 10 10 10	15 15 15 15 15		10 10 10 10 10	15 15 15 35 35	100-500 90-400 90-400 90-400 90-400	300 400 400 400 400	100 40 ² 40 ² 80 ² 80 ²	1.0 3.0 ² 3.0 ² 5.0 ² 5.0 ²	NS257 TO-1 TO-1 TO-1 TO-1	1,000 220 260 220 260
ACY11 ACY14 ACY17 ACY18 ACY19	P P P P P	32 32 70 50 50	16 16 12 12 12	30 30 32 30 30		5 5 6 6 6	12 12 10 10 10	38- 54- 50-150 40-120 80-250	10 10 300 300 300	35 35 40 40 40	0.4 0.4 1.0 1.0 1.3	TO-1 TO-1 TO-5 TO-5 TO-5	150 150 200 200 200
ACY20 ACY21 ACY22 ACY23 ACY27		40 40 20 32 40	12 12 12 16 30	20 20 15 30 20		6 6 6 30	10 10 10 12	50-145 90-250 30-300 50- 20-55	50 50 300 1 ¹ 1 ¹	40 40 40 0.5 40	1.0 1.3 1.0 1.0 ² 1.0 ²	TO-5 TO-5 TO-5 TO-1 TO-1	200 200 200 900 200
ACY28 ACY29 ACY30 ACY31 ACY32	P P P P P	40 40 40 40 32	30 30 40 20 16	15 15 20 30 30		30 30 30 12 10	12 12 12 5 10	45-150 45-150 60-200 35-70 50-150	1 ¹ 1 ¹ 1 ¹ 1 ¹ 1 ¹	40 40 40 40 27 ²	1.0 ² 1.0 ² 1.0 ² 1.0 ² 0.5	TO-1 TO-1 TO-1 TO-1 TO-1	200 200 200 200 900

¹hfe
²typical

CASE OUTLINE DRAWINGS & DIMENSIONS

Technical drawing of a mechanical part showing front and top views with dimensions.

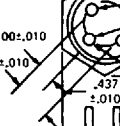
Front View Dimensions:

- Top flange width: .335
- Top flange thickness: .370
- Top flange inner width: .205
- Top flange outer width: .335
- Top flange height: .260
- Top flange base height: .240
- Top flange base width: 1.50 MIN
- Top flange base thickness: .008
- Top flange base width: .125
- Top flange base thickness: .019
- Top flange base width: .016

Top View Dimensions:

- Top flange width: .200
- Top flange thickness: .100
- Top flange inner width: .029
- Top flange outer width: .045
- Top flange base width: .034
- Top flange base thickness: .028
- Top flange base width: .028
- Top flange base thickness: .028
- Top flange base width: .028
- Top flange base thickness: .028

Technical drawing of a 3-lead chip carrier. The top view shows a circular body with a diameter of .650. Three leads extend from the bottom, with a center-to-center distance of .146 between adjacent leads and a lead thickness of .136. The leads are positioned at 90° intervals. The side view shows a maximum height of .115 and a body thickness of .330. The top view also indicates a .330 dimension from the center to the lead base. The side view shows a .524 dimension for the top width and a .444 dimension for the bottom width. The top view shows a .146 dimension from the center to the lead base and a .136 dimension for the lead thickness. The side view shows a .033 dimension for the top thickness and a .027 dimension for the bottom thickness. The side view also shows a .444 dimension for the top width and a .360 dimension for the bottom width.



Technical drawing of a hex head bolt. The drawing includes a side view at the top and an end view at the bottom. Dimensions and callouts are as follows:

- Side View Dimensions:**
 - UNDERCUT 1.0" $\pm .093 \pm .010$
 - Hex head height: $.500 \pm .015$ MAX.
 - Thread length: 1.500 MIN.
 - Thread pitch: 1/2 20 UNF 2A
 - Thread diameter: $.025 \pm .003$
 - Hex head width: $1.00 \pm .010$
- End View Dimensions:**
 - Hex head width: $.875$ MAX.
 - Hex head width: $.750 \pm .010$
 - Inner diameter: $.200 \pm .010$
 - Outer diameter: $.100 \pm .010$
- Callouts:**
 - A: Hex head
 - B: Thread
 - C: Thread relief
 - E: Undercut
- NOTE 1:**

THREAD RELIEF IS .090 MAX.
BY .430 DIA.
NOMINAL

Technical drawing of a transistor package showing side and top views with dimensions.

Side View Dimensions:

- Overall width: 1.250 DIA MAX.
- Base width: 1.005 DIA MAX.
- Base thickness: .210 MAX.
- Emitter thickness: .360 MAX.
- Lead thickness: .500 MAX.
- Lead width: .375 MAX.
- Lead height: .312 MAX.
- Lead diameter: .210 MAX.
- Lead pitch: .610 MAX.
- Lead thread: 10-32 UNF-2A
- Lead diameter: .137 MAX.
- Lead diameter: .123 MAX.

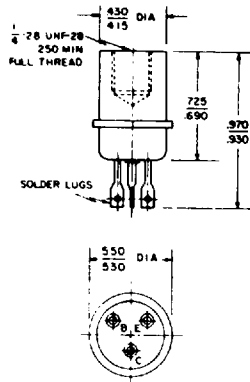
Top View Dimensions:

- Base diameter: .140 DIA. MAX.
- Base radius: .345 R
- Emitter diameter: .140 DIA. MAX.

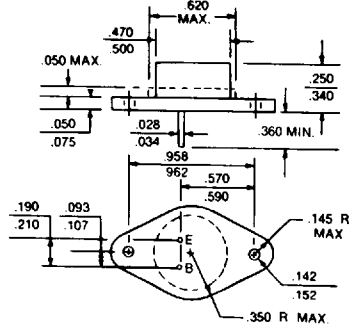
GERMANIUM POWER TRANSISTORS

CASE OUTLINE DRAWINGS & DIMENSIONS

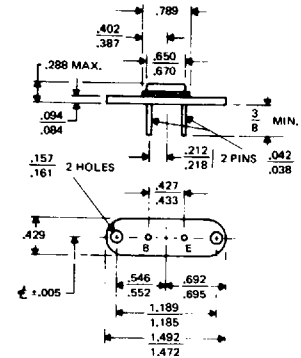
TO-13



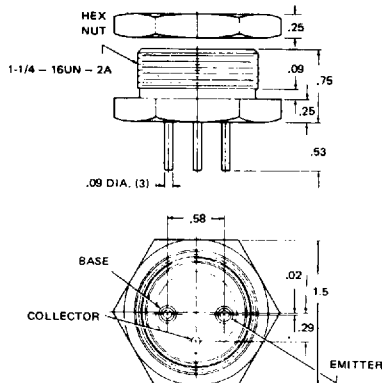
TO-66



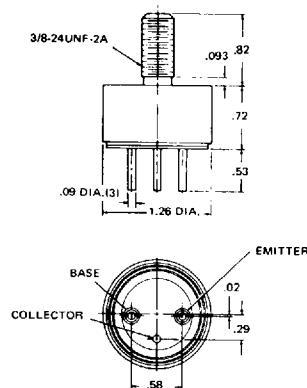
MS-7



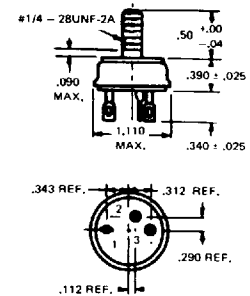
MT-22



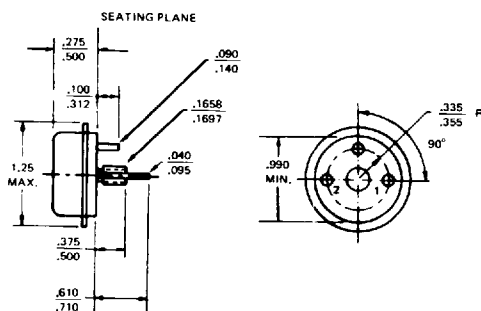
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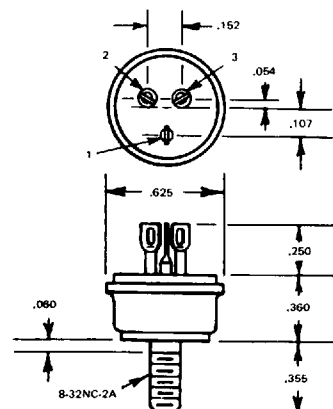
MT-7



TO-68



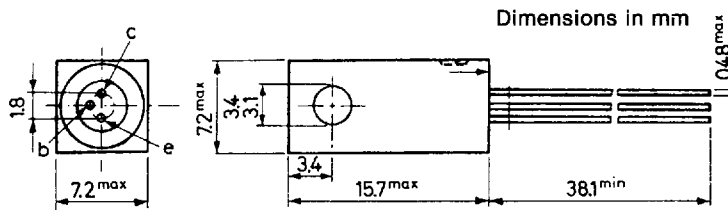
MT-36



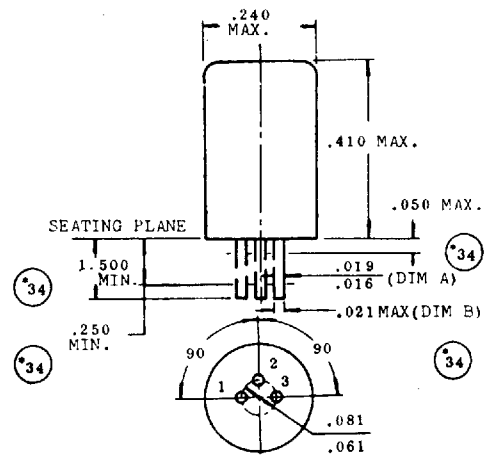
GERMANIUM POWER TRANSISTORS

CASE OUTLINE DRAWINGS & DIMENSIONS

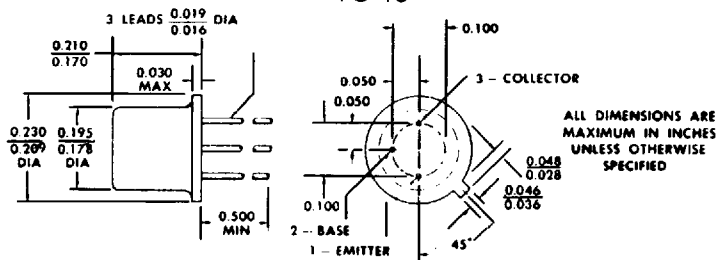
NS257



TO-1



TO-18



THE COLLECTOR IS ELECTRICAL CONTACT WITH THE CASE.

ALL JEDEC TO-18 DIMENSIONS AND NOTES ARE APPLICABLE.



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